Dual ordinal spectral ranking of journals and institutions

Docampo, D. and Cram, L.

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

Journal and institution ranking. The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval. Institutions fund individual researchers, implement training, oversee compliance, and maintain physical and information infrastructure. [Placeholder: overview of the joint ranking problem.]

Previous work on journal ranking. **?** introduced influence-per-reference weights for journal ranking; Vigna (2016) surveys the broader lineage; **?** traces the Katz–Hubbell–Pinski-Narin–PageRank chain. [Placeholder: expand journal ranking literature.]

How institutions are ranked. Institutional rankings, dominated by QS, Times Higher Education (THE), and ShanghaiRanking (ARWU), blend bibliometrics with reputational surveys, teaching metrics, and internationalisation. The Leiden Ranking counters with field-normalised bibliometrics, revealing niche leaders but lacking the reputational component of QS and THE. Spectral methods have not previously been applied jointly to sources and institutions. [Placeholder: expand institutional ranking literature.]

Theoretical framing. Following Wouters (1999), a *reference* written by a scientist becomes a *citation*—a directed edge (w_i, w_j)—only after indexer resolution to canonical work records.

We treat citations as structural relations within a functionally differentiated scientific system (**?**), setting aside authorial intent. Dual spectral rankings of journals and institutions treat citations as autopoietic communications within science's communicative system (Social System of Science Citation Theory, SSCT), where individual actors reside in the environment rather than constituting system elements (Tahamtan and Bornmann, 2022). This contrasts with actor-network theory (ANT), which foregrounds heterogeneous agents in assembling citation networks; ANT enters only in the Discussion. The resulting dual rankings are self-referential hierarchies that condition future citations. Our construction extends Breiger (1974): all four blocks of the joint attention matrix are primary flows, not derived projections.

Paper structure. Section 2 develops the mathematical framework. Section **??** describes the corpus construction and source selection. Section 4 presents results. Section 5 discusses validity and prospects.

2 Journal–institution citation networks

We begin corpus construction by selecting N_s sources $\mathfrak{S}$ in a field, discipline or category. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $[t_0^c, t_1^c]$ or the earlier contiguous target year t^t where $t^t = t_0^c - 1$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{(i,j) : w_i \in \mathfrak{W}^c,\, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of institutions $\mathfrak{U}'$ and a set of authors $\mathfrak{A}'$. Examining the frequency distributions of institutions and authors, we retain institutions with work counts $\geq \tau^U$ and authors with work counts $\geq \tau^A$, giving $\mathfrak{U}$ and $\mathfrak{A}$ respectively. Each work w_i has an *institution weight* under author-fractional counting,

$$\omega_{ik} \;=\; \sum_{j:\, k \in \sqcap_{ij}} \frac{1}{a_i}\,\frac{1}{u_{ij}}, \tag{1}$$

where a_i is the number of authors and u_{ij} the number of institutions of author j. Since ω_{iu} requires both an author and an institution, we drop any authorship where either the author $\notin \mathfrak{A}$ or the institution $\notin \mathfrak{U}$, and subsequently drop works that lose all their authorships. The institution weights are not renormalised to account for the omission of an authorship, so that $\sum_k \omega_{ik} \leq 1$.

When a work w_i references another work w_j we say that w_i gives attention to w_j. The attention given by a work can be either (a) the number of references, or (b) a fixed amount (Zitt and Small, 2008). Let $R_i = |\{j : (i,j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. We use the *reference normalisation weight* δ_i to select case (a) or case (b) respectively:

$$\delta_i \;=\; \begin{cases} 1 & \text{(full count)} \\ \bar{R}/R_i & \text{(fixed count).} \end{cases} \tag{2}$$

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \ldots, N_s$ and institutions indices $N_s + 1, \ldots, N$. A reference $(i,j) \in \mathfrak{R}$ gives attention across unit-type pairs: source–source (SS), source–institution (SI), institution–source (IS), and institution–institution (II).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_k = \mathbf{1}[w_i \in s_k]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (1). Attention flow between pairs of unit types is accumulated as four blocks over $\mathfrak{R}$ as rank-1 outer products,

$$\mathbf{C}_{XY} \;=\; \sum_{(i,j) \in \mathfrak{R}} \delta_i\, \mathbf{b}_i^X \left(\mathbf{b}_j^Y\right)^\top, \qquad XY \in \{SS, SI, IS, II\}, \tag{3}$$

with $\mathbf{b}_i^X$ weighting the referencing (citing) side and $\mathbf{b}_j^Y$ the referenced (cited) side.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \delta) \;=\; \begin{pmatrix} m_{SS}(1-\chi)^2\, \mathbf{C}_{SS} & m_{SI}\, \chi(1-\chi)\, \mathbf{C}_{SI} \\ m_{IS}\, \chi(1-\chi)\, \mathbf{C}_{IS} & m_{II}\, \chi^2\, \mathbf{C}_{II} \end{pmatrix}. \tag{4}$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0,1\}^4$, $m_{SI} = m_{IS}$ is bibliometrically meaningful for source-only $(1,0,0,0)$, institution-only $(0,0,0,1)$, bipartite $(0,1,1,0)$, and full-joint $(1,1,1,1)$ combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. We could name $\mathbf{C}$ the reference, attention or citation matrix and prefer the latter.

The model parameters are summarised in Table 1. Note that the corpus excludes (a) census work references not among target works, (b) non-census works that reference target works, and (c) authorships that include an omitted author or institution.

Spectral ranking. Let $\mathbf{r} = r_p = \sum_q C_{pq}$ be the total attention from unit p, $\mathbf{D}_r = \mathrm{diag}(r_p)$ the corresponding diagonal matrix, and $\mathbf{p}$ a boundary condition (Vigna, 2016) with entries $p_p = a_p / \sum_q a_q$, where $a_p = N_p^s$ for sources and $a_p = M_k$ for institutions. The row-stochastic citation matrix is $\mathbf{H} = \mathbf{D}_r^{-1}\mathbf{C}$.

Table 1: Model parameters $\boldsymbol{\theta}$.

Symbol	Name	Type	Role
Corpus construction			
$\mathfrak{S}$	source set	set	define the field, subject, category
$[t_0^c, t_1^c]$	census window	interval	works whose references are traced
$[t_0^t, t_1^t]$	target window	interval	works eligible to receive references
τ^U	institution threshold	integer	minimum work count to retain an institution
τ^A	author threshold	integer	minimum work count to retain an author
Matrix construction			
$\delta \in \{0,1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units; 1: a reference contributes 1 unit
$\mathbf{m} \in \{0,1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS, SI, IS, II)
$\chi \in [0,1]$	mixing weight	continuous	source–institution balance within each work
Ranking			
$\alpha \in [0,1)$	discount factor	continuous	downweight long walks; ensure convergence

The spectral ranking $\boldsymbol{\pi}$ (see Vigna, 2016, for an overview of the concept) is the largest eigenvector of the problem

$$\boldsymbol{\pi} \;=\; \alpha \mathbf{H}^\top \boldsymbol{\pi} + (1-\alpha)\mathbf{p}, \qquad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \tag{5}$$

with damping factor $\alpha \in [0,1)$. In the limit $\alpha \to 1$, equation (5) reduces to the eigenvector problem

$$\boldsymbol{\pi}^* = \mathbf{H}^\top \boldsymbol{\pi}^*, \qquad \mathbf{1}^\top \boldsymbol{\pi}^* = 1, \quad \boldsymbol{\pi}^* \geq 0, \tag{6}$$

whose solution is unique when $\mathbf{H}$ is primitive. In that limit, Geller (1978) notes that $\boldsymbol{\pi}^*$ is the limiting stationary distribution of a irreducible, aperiodic Markov chain with transition matrix $\mathbf{H}$. She shows that $\boldsymbol{\pi}^*$ is the *influence weight* defined by Pinski and Narin (1976), scaled by the reference count proportion $\mathbf{r}/|\mathbf{r}|$.

The spectral ranking $\boldsymbol{\pi}$ depends on the number of works in each unit, while

$$\mathbf{v} \;=\; \mathbf{A}^{-1}\boldsymbol{\pi}, \qquad \mathbf{A} = \mathrm{diag}(p_r) \tag{7}$$

removes this dependence. Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is a ranking that is uniquely invariant to several reasonable constraints (but see Serrano, 2004).

Computation via power iteration. To compute $\boldsymbol{\pi}^*$ we define the modified matrix

$$\mathbf{H}^* \;=\; \alpha \mathbf{H} + (1-\alpha)\mathbf{p}\mathbf{1}^\top, \tag{8}$$

so that the fixed-point condition $\mathbf{H}^{*\top}\boldsymbol{\pi}^* = \boldsymbol{\pi}^*$ is equivalent to equation (5). Because $\mathbf{p} > 0$ and $0 < \alpha < 1$, every entry of $\mathbf{H}^*$ is strictly positive, making $\mathbf{H}^*$ a primitive row-stochastic matrix. By the Perron–Frobenius theorem, $\mathbf{H}^{*\top}$ therefore has a unique stationary distribution $\boldsymbol{\pi}^*$, recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} \;=\; \mathbf{H}^{*\top}\boldsymbol{\pi}^{(k)}, \tag{9}$$

converging to $\boldsymbol{\pi}^*$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$. We report $\mathbf{v}^*$ over $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$. Convergence analysis is in the Supplement.

Computation. $\mathbf{C}(\chi, \mathbf{m}, \rho)$ is sparse (nnz $\leq 4|\mathcal{R}|$) and is built from a weighted edge list $\mathcal{E}$ accumulated over $\mathcal{R}$ with per-reference weight $\delta_i(\rho)$, stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathcal{R} \to \mathcal{E} \to \mathbf{C} \to \mathbf{H} \to \boldsymbol{\pi}^* \to \mathbf{v}^*$ and its parameter entry points are detailed in the Supplement. Implementation uses `scipy.sparse.csr_matrix`.

3 Source Selection

We require each source in the corpus to satisfy two conditions: (i) a verified OpenAlex source identifier (OAS ID), providing a stable anchor for work extraction; and (ii) disciplinary evidence from at least one of four complementary registers confirming that the source publishes economics or business research. The four registers differ in how they define disciplinary scope and in the population of journals they cover; together they cast a wide but disciplined net.

3.1 Four Disciplinary Registers

JQL. Harzing's *Journal Quality List* **?** assembles peer assessments of journal quality across business and economics disciplines. Its scope is already confined to the target fields; no further filtering is applied. The 71st edition contains 827 journals.

SJL. The *ERA 2023 Submission Journal List* **?**, maintained by the Australian Research Council, assigns up to three *Field of Research* (FoR) codes to each journal. SJL entries are retained where any of the three assigned FoR codes falls within Division 35 (*Commerce, Management, Tourism and Services*) or Division 38 (*Economics*), encompassing both two-digit divisional codes and all four-digit group codes (3501–3509, 3801–3803, 3899). Using any FoR column rather than only the primary FoR 1 captures cross-disciplinary journals classified primarily in adjacent fields that nonetheless carry a secondary economics or management classification. After filtering, 2,469 journals remain.

MJL. The *Clarivate Master Journal List* **?**, drawn from the Web of Science Journal Citation Reports, assigns each journal a Web of Science category. Entries are retained where that category is one of: ECONOMICS, TRANSPORTATION, MANAGEMENT, BUSINESS, or BUSINESS, FINANCE. After filtering, 1,413 journals remain.

OJL. The *OpenAlex Journal List* (OJL) is constructed directly from the OpenAlex topics taxonomy. Each source record in the OpenAlex snapshot carries topic assignments derived from the content of its indexed works, with each topic linked to a *subfield*, *field*, and *domain*. The OJL retains all OpenAlex sources for which at least one indexed work carries a topic belonging to Field 14 (*Economics, Econometrics and Finance*) or Field 20 (*Business and Management*), yielding 42,756 source entities. For each source, the OJL records the *in-scope ratio*—the share of indexed works whose primary topic falls within Fields 14 or 20—as a measure of disciplinary concentration. Unlike the three register-based sources, OJL provides content-derived evidence rather than editorial classification, and is the only register that guarantees an OAS ID by construction.

3.2 Joining to OpenAlex Sources

Each of JQL, SJL, and MJL is joined independently to the OpenAlex sources snapshot using ISSN as the matching key. For JQL, the single ISSN column is matched against the OpenAlex preferred ISSN (`issn_l`) and all alternative ISSNs in the `issn` field. For MJL, both the print and electronic ISSN fields are tried in sequence. For SJL, all seven ISSN columns (`ISSN 1` through `ISSN 7`) are tried in sequence, stopping at the first hit. OJL sources carry OAS IDs natively and require no ISSN matching.

The four matched sets are unioned, deduplicating by OAS ID, to produce the *long-list* of 2,751 candidate sources. Match statistics for the three register-based sources are shown in Table 2; the 462 long-list entries with no OJL match are examined in Section 3.3.

Match rates are high for JQL (99.9%) and MJL (98.6%) but more modest for SJL (88.8%), reflecting the ERA register's deliberate inclusion of lower-profile national and regional journals. Of the 276 unmatched SJL entries, 132 are small specialist journals not indexed in OpenAlex, 47 are probable ISSN mismatches (held for manual review), and the remainder are regional, non-Anglophone, tax/legal specialist, or newsletter/working-paper series entries (Table 3). The 20 unmatched MJL entries are predominantly regional or non-Anglophone publications; the single unmatched JQL entry is the *B.E. Journal of Economic Analysis and Policy*, absent from the OpenAlex sources snapshot.

Table 2: Matching of the three register-based sources to OpenAlex source identifiers (OAS IDs). Pairwise overlaps are based on shared OAS IDs among matched entries.

Register	**Journals**	**Matched**	**Match rate**	**Unmatched**
JQL	827	826	99.9%	1
MJL	1,413	1,393	98.6%	20
SJL	2,469	2,193	88.8%	276
Pairwise overlap of matched OAS				
MJL ∩ SJL	1,008 sources			
MJL ∩ JQL	540 sources			
SJL ∩ JQL	618 sources			
MJL ∩ SJL ∩ JQL	505 sources			
Union (long-list)	**2,751 unique OAS**			

Table 3: Categorisation of the 276 SJL journals not matched to an OpenAlex source ID.

Category	**Count**
Small specialist journal (not indexed in OA)	132
Possible ISSN mismatch (name match found in OA)	47
Regional / national journal	30
Non-English / regional language journal	22
Tax / legal specialist journal	14
Education / pedagogical journal	12
Newsletter / trade publication	9
Working paper / discussion paper series	6
Actuarial / insurance journal	3
Annual edition / book series	1
Total	**276**

3.3 Disciplinary Scope Filtering

The three register-based sources cast a deliberately broad net. The ERA SJL captures any journal with a business or economics secondary classification; the MJL includes transportation and management science outlets; Harzing's JQL includes journals that, while useful references for management researchers, are primarily housed in adjacent disciplines. The OJL cross-check provides content-derived evidence to adjudicate these borderline cases.

Of the 2,751 long-list OAS, 2,289 (83.2%) are found in the OJL, confirming independent content-based recognition as an economics or business source. The remaining 462 OAS (16.8%) are absent from the OJL—OpenAlex has not classified any of their indexed works under Fields 14 or 20. This arises for two distinct reasons: genuine disciplinary mismatch, or a gap in the OpenAlex topic classifier. The 82 long-list OAS carrying a MJL category of Economics, Management, Business, or Business, Finance but absent from the OJL exemplify the second case: they are confirmed in-scope by the Web of Science system but classified by OpenAlex primarily under transport science or operations research (e.g. *Transportation Research* parts B, D, and F). These 82 sources are retained on the strength of their WoS category.

Each of the remaining 380 OAS absent from the OJL is assigned a scope signal based on register membership and journal name (Table 4). Two activity criteria are applied uniformly across all 2,751 long-list OAS: sources with no indexed publication after 1999 (*Inactive pre-2000*) and sources with fewer than one OA-indexed work per active year (*Very low output*) are excluded as they cannot contribute meaningfully to a works-based corpus regardless of disciplinary classification.

Table 4: Scope signal assigned to the 462 long-list OAS absent from the OJL, based on register membership and journal name.

Scope signal	**Count**	**Interpretation**
WoS-confirmed scope	82	MJL category confirms scope; OJL absence reflects classifier gap. Retained.
JQL+SJL borderline	36	In both Harzing and ERA; names ambiguous. Retained pending review.
JQL peripheral	79	JQL only; names suggest adjacent discipline (statistics, IS, sociology). Excluded.
Adjacent field	105	Name clearly outside target fields (natural sciences, medicine, political science, psychology). Excluded.
SJL adjacent	186	SJL only; secondary FoR in transport, logistics, or engineering. Excluded.
Total	**462**	

3.4 Final Source List

Table 5 summarises the scope flag applied to each of the 2,751 long-list OAS, with flags assigned in priority order: (i) inactivity before 2000, (ii) very low output, (iii) scope signal for OJL-absent sources, (iv) in-scope otherwise.

Table 5: Distribution of scope flags across the 2,751 long-list OAS. Sources flagged *In scope* or *WoS-confirmed* are retained for work extraction.

Flag	**Count**	**Share**	**Basis**
In scope	2,259	82.1%	OJL-matched; active; adequate OA output
WoS-confirmed scope	82	3.0%	MJL category; OJL absence is classifier gap
Excluded:			
SJL adjacent	209	7.6%	SJL-only; transport / logistics / engineering
JQL peripheral	79	2.9%	JQL-only; adjacent social or formal sciences
Adjacent field	59	2.1%	Name-identified non-economics outlet
JQL+SJL borderline	33	1.2%	Borderline; retained for manual review
Very low output	25	0.9%	<1 OA-indexed work per active year
Inactive pre-2000	5	0.2%	No publication after 1999
In scope (net)	**2,341**	**85.1%**	
Excluded	**410**	**14.9%**	
Total	**2,751**	**100%**	

The 2,341 in-scope OAS form the *scope-filtered source list* used for work extraction. The 505 sources confirmed by all four registers (JQL ∩ SJL ∩ MJL ∩ OJL) represent the highest-consensus core: outlets simultaneously recognised by an editorial quality list, the ERA classification, the WoS category system, and OpenAlex content analysis. The 33 JQL+SJL borderline sources are included pending manual review; the 25 very-low-output sources are a practical rather than disciplinary exclusion.

4 Results

Baseline model. [Placeholder.]

Parameter sensitivity. [Placeholder: $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$; $\chi \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$; $\mathbf{m} \in \{(1,0,0,0), (0,1,1,0), (1,1,1,1)\}$; $\rho \in \{0,1\}$; τ sweep.]

Bootstraps. [Placeholder.]

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Prospects

[Placeholder.]

8 Addenda

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